

YEAR 11 YEARLY EXAMINATION

SEPTEMBER 2011

- Working time allowed – 120 minutes.
- Write using black or blue pen.
- Approved calculators may be used.
- All necessary working should be shown.
- Start each question on a new page.
- Attempt all questions.
- Questions are of equal value
- Full marks may not be awarded if working is poorly set out or difficult to read

NAME: _____ TEACHER: _____

[illegible]

Question 1**Marks**

a) Solve $\frac{1}{4}(2x-3) + x = 2$

/2

b) Simplify $\frac{x+2}{2} - \frac{x-1}{3}$

/2

- c) Divide the product of
 2.9384×10^{12} and 1.58×10^{11} by 2.068×10^{10}
Give your answer correct to 3 significant figures

/2

- d) Find the values of a and b if

$$\frac{3}{\sqrt{5+2}} = a + \sqrt{b}$$

/2

- e) Brian pays \$34.50 for a shirt which was originally marked at \$45.00
Calculate the percentage discount correct to one decimal place

/2

- f) Find the exact value of $\tan 300^\circ$

/1

Question 2**Marks**

a) Find the range of values of p for which $p - 6x - x^2$ is negative definite /2

b) Use the general quadratic formula to solve: $2x^2 - 2x - 5 = 0$

Leave your answer in simplified surd form. /2

c) (i) For the function: $f(x) = x^2 + 2$ write down the simplified expression for /2

$$\frac{f(x+h) - f(x)}{h}$$

(ii) Hence or otherwise find: /1

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

d) Evaluate: /2

$$\lim_{x \rightarrow 3} \frac{x^2 - 2x - 3}{x - 3}$$

e) Sketch the function $y = |x| - 3$ showing all important features including intercepts /2

Question 3**Marks**

a) Solve: $|2x + 3| = 7$

/2

b) Differentiate with respect to x

(i) $y = (3x - 5)^4$

/2

(ii) $y = \frac{2x-3}{x+5}$

/2

(iii) $y = 2x^2(5x + 2)^3$

/2

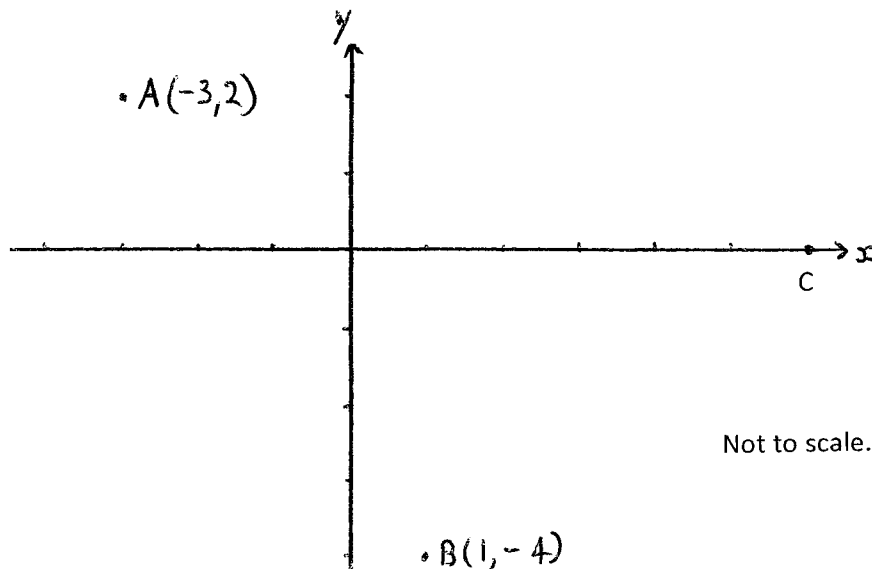
c) Find the equation of the normal to the curve $y = 3x^2 - 5x + 2$ at the point $(1, 0)$

/3

Question 4

Marks

a)



A, B and C are the vertices of a right angled triangle with $\angle ABC = 90^\circ$

(i) Find the length AB (exact value)

/1

(ii) Find the gradient of AB

/1

(iii) Find the equation of BC in general form.

/2

(iv) Find the coordinates of C

/1

(v) Find the area of $\triangle ABC$

/2

b) Find the shortest distance between the parallel lines

/2

$3x+2y+1=0$ and $3x+2y-1=0$? (leave answer as a surd)

c) Solve $|2x - 1| = x - 7$

/2

Question 5**Marks**a) Solve for $0 \leq \theta \leq 360^\circ$

(i) $2\cos \theta + 1 = 0$

/2

(ii) $\operatorname{cosec}^2 \theta = 2$

/2

b) Find the quadratic equation whose sum of roots is 5 and product of roots is -3

/2

c) Find the values of k in $3x^2 - kx + 3 = 0$ if the quadratic equation has no real roots.

/2

d) If α and β are the roots of the quadratic equation $x^2 + 3x - 5 = 0$, find

(i) $\alpha + \beta$

/1

(ii) $\alpha\beta$

/1

(iii) $\alpha^2 + \beta^2$

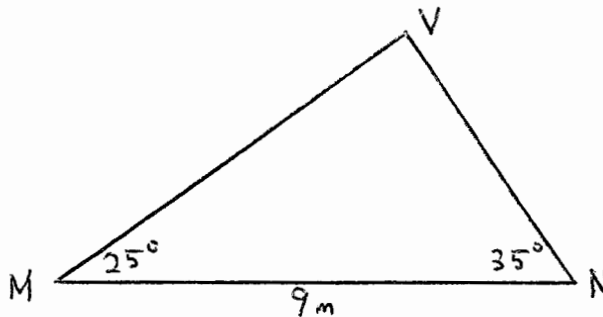
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Question 6**Marks**

a) Prove that $\frac{1}{\cot^2 \theta + 1} = (1 - \cos \theta)(1 + \cos \theta)$

/3

b) The framework of a roof MNV is shown in the diagram below



The frame spans a horizontal distance of 9m with beams VM and VN inclined at angles 25° and 35° respectively to the horizontal.

i. Calculate the length of the beam VN to the nearest centimetre.

/2

ii. Hence or otherwise find the area enclosed by the frame (correct to 1 decimal place)

/2

c) Draw a diagram showing the region described by the following inequalities:

/2

$$y \leq 4 - x^2 \quad y \geq 0 \quad x \leq 0$$

d) Solve

$$2^{3x-1} = \frac{1}{4}$$

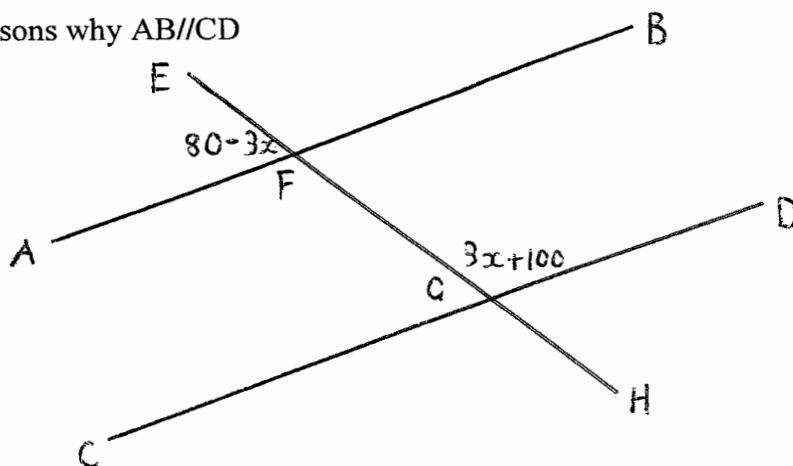
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Question 7

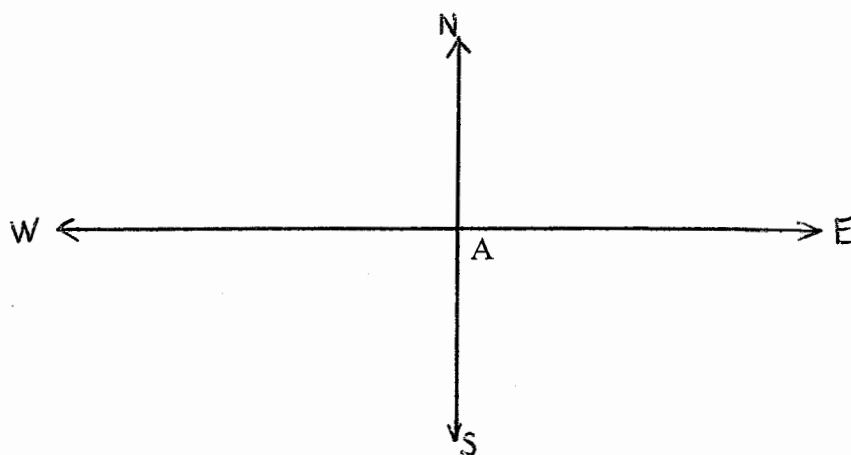
Marks

- a) Give reasons why $AB \parallel CD$

/2



- b) From town A a plane flies on a bearing of 310° for 16km to town B. It then changes direction and flies at a bearing of 200° for 20km to town C.



- i. Copy the diagram above including the given information

/1

- ii. Find the distance of town A from town C (correct to 2 decimal places)

/2

- c) Find the x coordinate of the point on the curve $y = x^2 + 2$ where the tangent has a gradient of -2 . /1

d) For the function
$$f(x) = \begin{cases} x^3 + 1 & \text{if } x > 2 \\ 2x & \text{if } -1 \leq x \leq 2 \\ 5 & \text{if } x < -1 \end{cases}$$

Find $f(-3) - f(5) + f(2)$ /2

e) Solve $25 - n^2 \geq 0$ /2

f) Find the angle sum of a 16 sided polygon. /1

Question 8**Marks**

a) Solve $(2^x)^2 - 6(2^x) + 8 = 0$

/2

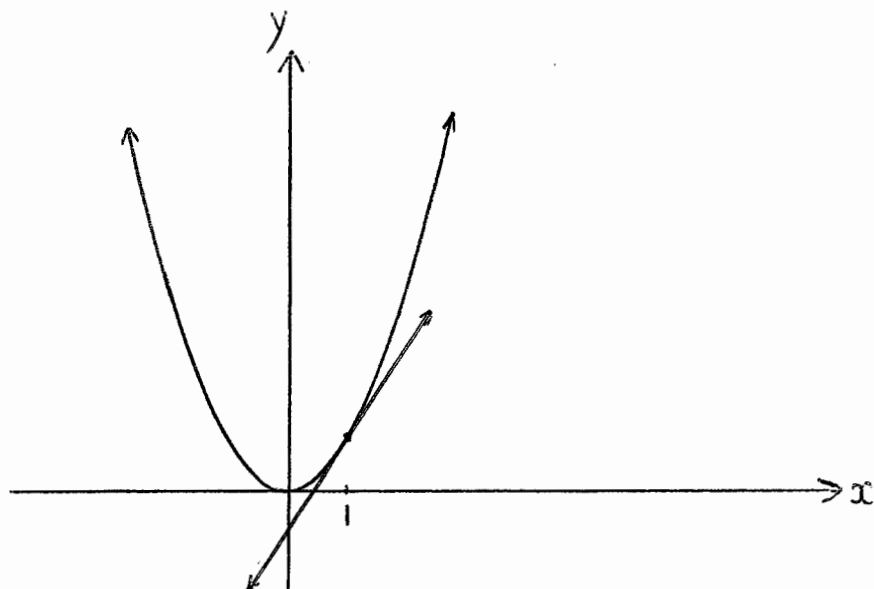
b) (i) State the domain and range of $y = \frac{1}{x-1} + 1$

/2

(ii) Sketch the curve showing all important features including intercept

/2

c)



i. Find the equation of the tangent to $y = x^2$ at $x = 1$ /2

ii. Find where this tangent crosses the x and y axes. /2

iii. Find the area of the triangle enclosed by the tangent and the x and y axes. /1

Teacher's Name:

Student's Name/N°:

Solutions 2011 2 Unit Preliminary Final

Question 1

a) $\frac{1}{4}(2x-3) + x = 2$

$$2x-3 + 4x = 8$$

$$6x-3 = 8$$

$$6x = 11$$

$$x = \frac{11}{6}$$

b) $\frac{x+2}{2} - \frac{x-1}{3}$

$$\frac{3(x+2) - 2(x-1)}{6}$$

$$\frac{3x+6-2x+2}{6}$$

$$\frac{x+8}{6}$$

c) 2.25×10^{13}

d) $\frac{3}{\sqrt{5}+2} = a + \sqrt{b}$

$$\frac{3}{\sqrt{5}+2} \times \frac{\sqrt{5}-2}{\sqrt{5}-2}$$

$$= \frac{3\sqrt{5}-6}{1}$$

$$= -6 + \sqrt{45}$$

$$a = -6 \quad b = 45$$

e) Discount = \$10.50

$$\frac{10.50}{45} \times \frac{100}{1}$$

$$= 23.3\%$$

f) $-\tan 60$

$$= -\frac{\sqrt{3}}{1}$$

$$= -\sqrt{3}$$

Question 2

a) $-x^2 - 6x + p$

$$\Delta < 0$$

$$36 - 4 \times 1 \times p < 0$$

$$36 + 4p < 0$$

$$4p < -36$$

$$p < -9$$

b) $2x^2 - 2x - 5 = 0$

$$\frac{2 \pm \sqrt{(-2)^2 - 4 \times 2 \times (-5)}}{2 \times 2}$$

$$x = \frac{2 \pm \sqrt{44}}{4}$$

$$x = \frac{2 \pm \sqrt{44}}{4}$$

$$= \frac{2 \pm 2\sqrt{11}}{4}$$

$$x = \frac{1 \pm \sqrt{11}}{2}$$

c) (i) $\frac{f(x+h) - f(x)}{h}$

$$= \frac{(x+h)^2 + 2 - (x^2 + 2)}{h}$$

$$= \frac{x^2 + 2xh + h^2 + 2 - x^2 - 2}{h}$$

$$= 2x + h$$

(ii) $\lim_{h \rightarrow 0} 2x + h$

$$= 2x$$

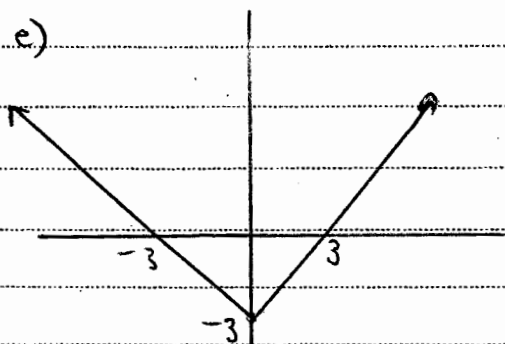
Teacher's Name:

Student's Name/N^o:

$$d) \lim_{x \rightarrow 3} \frac{(x+1)(x-3)}{(x-3)}$$

$$= 4$$

e)

Question 3

$$a) 2x+3=7 \quad \text{or} \quad 2x+3=-7$$

$$2x=+4 \quad \text{or} \quad 2x=-10$$

$$x=+2 \quad \text{or} \quad -5$$

$$b) \frac{dy}{dx} = 4(3x-5)^3 \times 3$$

$$= 12(3x-5)^3$$

$$c) \frac{dy}{dx} = \frac{(x+5) \times 2 - (2x-3) \times 1}{(x+5)^2}$$

$$= \frac{2x+10-2x+3}{(x+5)^2}$$

$$= \frac{13}{(x+5)^2}$$

$$c) \frac{dy}{dx} = 2x^2 \times 3(5x+2)^2 \times 5$$

$$+ 4x(5x+2)^3$$

$$= 30x^2(5x+2)^2 + 4x(5x+2)^3$$

$$e) y = 3x^2 - 5x + 2$$

$$\frac{dy}{dx} = 6x - 5$$

$$\text{At } x=1, \quad \frac{dy}{dx} = 1$$

$$\therefore m_{\text{normal}} = -1$$

$$y - 0 = -1(x - 1)$$

$$y = -x + 1$$

Question 4

$$a) \text{ci) } AB = \sqrt{(-3-1)^2 + (2-4)^2}$$

$$= \sqrt{16+36}$$

$$= \sqrt{52} \quad \text{or} \quad 2\sqrt{13}$$

$$\text{cii) } m_{AB} = \frac{2-4}{-3-1}$$

$$= \frac{6}{-4} = -\frac{3}{2}$$

$$\text{ciii) } m_{BC} = \frac{2}{3}$$

$$y - -4 = \frac{2}{3}(x - 1)$$

$$3y + 12 = 2x - 2$$

$$0 = 2x - 3y - 14$$

$$\text{civ) } (7, 0) \quad \text{or} \quad x=7$$

$$\text{cv) } d_{BC} = \sqrt{(7-1)^2 + (0-4)^2}$$

$$= \sqrt{52}$$

$$A_{ABC} = \frac{1}{2} \times \sqrt{52} \times \sqrt{52} = 26$$

Teacher's Name:

Student's Name/N^o:b) $(0, -\frac{1}{2})$ lies on $3x + 2y + 1 = 0$

$$d = \frac{|3 \times 0 + 2 \times -\frac{1}{2} + 1|}{\sqrt{3^2 + 2^2}}$$

$$= \frac{2}{\sqrt{13}} \text{ or } \frac{2\sqrt{13}}{13}$$

c) $|2x - 1| = x - 7$

$2x - 1 = x - 7 \text{ or } 2x - 1 = -(x - 7)$

$x = -6 \text{ or } 3x = 8$

$x = \frac{8}{3}$

$\therefore$ No solution as RHS < 0 for both answers

Question 5

a) i) $2\cos\theta + 1 = 0$

$\cos\theta = -\frac{1}{2}$

$\theta = 120^\circ, 240^\circ$

ii) $\operatorname{cosec}^2\theta = 2$

$\sin^2\theta = \frac{1}{2}$

$\sin\theta = \pm \frac{1}{\sqrt{2}}$

$\theta = 45^\circ, 135^\circ, 225^\circ, 315^\circ$

b) $x^2 - (-\frac{b}{a})x + \frac{c}{a} = 0$

$x^2 - 5x - 3 = 0$

c) $\Delta < 0$

$k^2 - 4 \times 3 \times 3 < 0$

$k^2 - 36 < 0$

$-6 < k < 6$

d) i) $-\frac{b}{a} = -3$

ii) $\frac{c}{a} = -5$

iii) $(\alpha + \beta)^2 - 2\alpha\beta$
 $(-3)^2 - 2 \times -5$
 19

Question 6

a) $\frac{1}{\cot^2\theta + 1}$

$= (1 - \cos\theta)(1 + \cos\theta)$

$= 1 - \cos^2\theta$

$= \sin^2\theta$

$= \frac{1}{\operatorname{cosec}^2\theta}$

$= \frac{1}{\cot^2\theta + 1}$

$= \text{LHS}$

Teacher's Name:

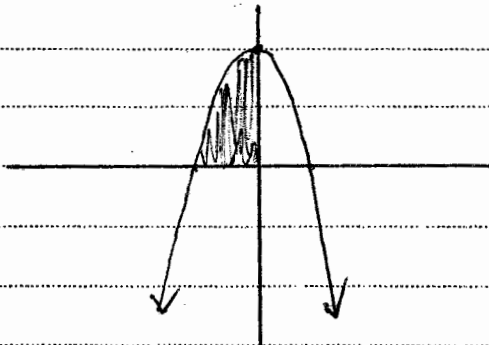
Student's Name/N^o:

$$b) \text{ci) } \frac{VN}{\sin 25} = \frac{9}{\sin 120}$$

$$\underline{VN = 4.39m}$$

$$\begin{aligned} \text{cii) } A &= \frac{1}{2} ab \sin C \\ &= \frac{1}{2} \times 9 \times 4.39 \times \sin 35 \\ &= \underline{11.3 m^2} \end{aligned}$$

c)



$$\begin{aligned} d) \quad 2^{3x-1} &= \frac{1}{4} \\ 2^{3x-1} &= 2^{-2} \\ 3x-1 &= -2 \\ 3x &= -1 \\ x &= \underline{-\frac{1}{3}} \end{aligned}$$

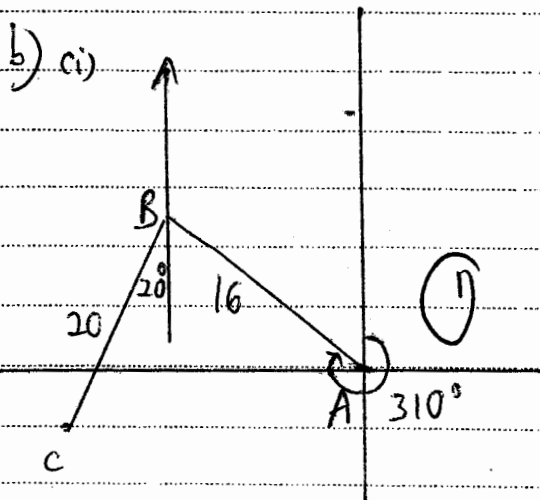
Question 7

$$a) \angle BFC = 80 - 3x \quad (\text{vertically opposite}) \quad (1)$$

$$\begin{aligned} \angle BFC + \angle FGD \\ &= 80 - 3x + 3x + 100 \\ &= 180^\circ \end{aligned}$$

$\therefore AB \parallel CD$ (co-interior angles supplementary) (1)

b) ci)



$$\begin{aligned} \text{cii) } AC^2 &= 20^2 + 16^2 - 2 \times 20 \times 16 \times \cos 70 \\ AC &= \underline{20.91 km} \quad (2) \end{aligned}$$

$$\begin{aligned} a) \quad \frac{dy}{dx} &= 2x = -2 \\ \text{at } x &= -1 \quad (1) \end{aligned}$$

$$\begin{aligned} d) \quad 5 - (5^3 + 1) + 2 \times 2 & \quad (1) \\ &= 5 - (126) + 4 \\ &= \underline{-117} \quad (1) \end{aligned}$$

$$\begin{aligned} e) \quad 25 - n^2 &\geq 0 \\ (5-n)(5+n) &\geq 0 \\ n-5 &\leq n \leq 5 \end{aligned}$$

(2)

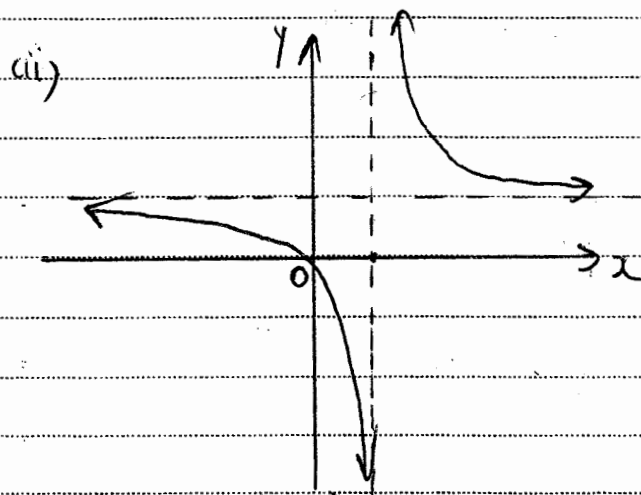
$$\begin{aligned} f) \quad (n-2) \times 180 \\ 14 \times 180 \\ &= \underline{2520^\circ} \quad (1) \end{aligned}$$

Teacher's Name: _____

Student's Name/N^o: _____Question 8

a) Let $u = 2^x$
 $u^2 - 6u + 8 = 0$
 $(u - 4)(u - 2) = 0$
 $u = 4$ or 2
 $2^x = 4$ or $2^x = 2$
 $x = 1$ or 2

b) i) D : All x except $x = 1$
 R : All y except $y = 1$



c) i) $\frac{dy}{dx} = 2x$
 At $x = 1$

ii) $(0, -1)$ $(\frac{1}{2}, 0)$

$m = 2$ $(1, 1)$
 $y - 1 = 2(x - 1)$
 $y = 2x - 1$

cii) $A = \frac{1}{2} \times 1 \times \frac{1}{2}$
 $= \frac{1}{4} \text{ units}^2$